

Bending-induced birefringence in single-mode fibers

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Received February 19, 1980

In a bent optical fiber, lateral internal stress induces a birefringence, β_b . At $\lambda = 0.633 \mu\text{m}$ in a silica fiber, $\beta_b = 7.7 \times 10^7 r^2/R^2 \text{ deg/m}$, where r is the fiber radius and R is the bending radius.

Birefringence in single-mode optical fibers is known to result (1) from deviations of the core geometry from the ideal straight circular cylinder and (2) from mechanical stress through the elasto-optic effect. Thus various kinds of birefringence have recently been explained, occurring in fibers with elliptical core,¹ in fibers under lateral pressure,² and in twisted fibers.³ In all cases, first-order electromagnetic perturbation theory and the well-known first-order elastic stress fields give satisfactory solutions. For bending-induced birefringence β_b however, no such simple explanation is possible. Several attempts to explain β_b gave results too small by orders of magnitude.⁴

In this Letter, we point out that bending-induced birefringence is essentially a stress effect. It results from the lateral, compressive stress $-\sigma_x$ that builds up in a bent fiber under the conditions of "large" deformations. The well-known dominant stress component⁵ in a bent fiber (Fig. 1) is $\sigma_z = \kappa Ex$, where $\kappa = 1/R$ is the curvature and E is Young's modulus. Across the xy plane, σ_z is a tensile stress in outer ($x > 0$) layers, but σ_z is compressive in all inner ($x < 0$) layers. Consequently, the outer layers exert a pressure $-\sigma_x$ in the radial (R) direction on the inner layers. This pressure is highest on the fiber axis and causes the birefringence, much like the birefringence caused by an external lateral force.²

The dominant stress σ_z does not contribute directly to the birefringence: No stress effect proportional to κ can exist because σ_z is an odd function of x , whereas the electromagnetic wave functions of the two orthogonal linearly polarized HE_{11} modes are even functions. Therefore, all first-order perturbation integrals³ vanish. Moreover, no geometry effect proportional to κ exists, because in that order of approximation the core deformation has a *fourfold* symmetry.

It is not surprising, therefore, that a second-order dependence, $\beta_b \sim \kappa^2$, has been observed.⁴ A full theoretical analysis would require second-order electromagnetic perturbation theory as well as the evaluation of the elastic stress up to terms $\sim \kappa^2$. Of the large number of terms in such an analysis, our present study concerns only one. As we find it in good agreement with our measurements, we believe all other terms to be much smaller.

For a calculation of the lateral stress in the cross section $z = 0$, we postulate the fiber as elastically homogeneous and isotropic, and we consider the limit of

weak curvature, $\kappa r \ll 1$, where $2r$ is the outer diameter of the fiber. The layer dx in Fig. 1 (b) is bent at a radius $(R + x)$. Elementary mechanics shows that the tension $\sigma_x(x)$ in this layer contributes for $|\sigma_x| \ll |\sigma_z|$ with

$$\sigma_x(x + dx) - \sigma_x(x) = (R + x)^{-1} \sigma_z(x) dx \quad (1)$$

to the lateral stress component σ_x . Here we may neglect $|x| \ll R$ and obtain

$$\partial \sigma_x / \partial x = \kappa^2 Ex. \quad (2)$$

Comparing this relation with the fundamental equations of elasticity,⁵ we recognize that $-\kappa^2 Ex$ plays the role of a compressive volume force density. For a simple estimate of the resulting birefringence, we ignore all other stress components ($\sigma_y = 0$) and integrate Eq. (2) along the x axis with $\sigma_x(\pm r) = 0$ at the fiber surfaces,

$$\sigma_x(x) = \kappa^2 (E/2)(x^2 - r^2). \quad (3)$$

This stress modifies the refractive index of the fiber material. In a weakly guiding single-mode fiber of core

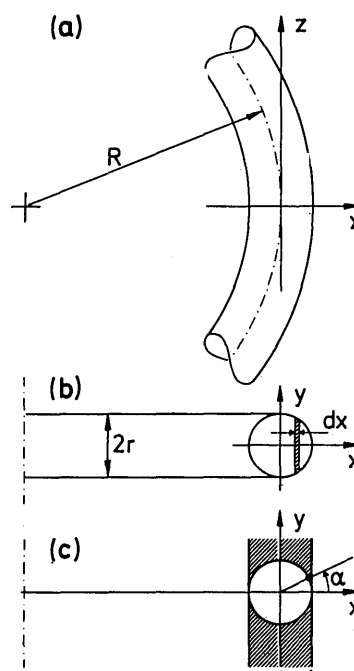


Fig. 1. Geometry of a bent fiber.

index n , the birefringence becomes $\beta_b \equiv k_x - k_y \approx k(\delta n_x - \delta n_y)$, where k_x and k_y are the propagation constants of the HE_{11} mode polarized along the x and y axes, respectively, and $k = 2\pi/\lambda_{vac}$. The elasto-optic index changes δn_x and δn_y are evaluated from the stress $\sigma_x(0)$ at the fiber axis by⁶ $\delta n_i = -(n^3/2)\Sigma p_{ij}\epsilon_j$. The p_{ij} denote the strain-optical coefficients, and the ϵ_j are the components of strain. Using Poisson's ratio ν , we have $\epsilon_x - \epsilon_y = (\sigma_x - \sigma_y)(1 + \nu)/E$, and we arrive at

$$\beta_b = 0.25kn^3(p_{11} - p_{12})(1 + \nu)\kappa^2r^2 \quad (4)$$

The rigorous integration of Eq. (2) given in the Appendix happens to give exactly the same result. Equation (4) predicts correctly the observed⁴ variation $\beta_b \sim \kappa^2$. Moreover, it shows that β_b is proportional to the square of the outer fiber diameter $2r$. Our measurements on three fibers of different diameters confirm this dependence (Fig. 2).

These measurements were performed at ~ 1 - 2 -m-long sections of weakly birefringent single-mode fibers from various manufacturers. To exclude any other stress effects in the measurements, the fibers were laid out in one or more circular turns on a flat surface and held down only with high-viscosity oil. He-Ne laser light was coupled into one fiber end at 45° azimuth relative to the plane, and the state of polarization at the output end was analyzed with a Soleil-Babinet compensator and Glan-Thompson analyzer.

Inserting into Eq. (4) the published⁷ material constants of fused silica [$n = 1.46$; $\nu = 0.17$; $(p_{11} - p_{12}) = -0.15$], and with $\lambda = 0.633 \mu\text{m}$, we obtain

$$\beta_b = -7.7 \times 10^7 \kappa^2 r^2 \text{ deg/m}. \quad (5)$$

This result should hold universally for weakly guiding silica fibers, regardless of their core diameters and index profiles. The theoretical prediction is supported by the good coincidence of our measurements (Fig. 2) with the

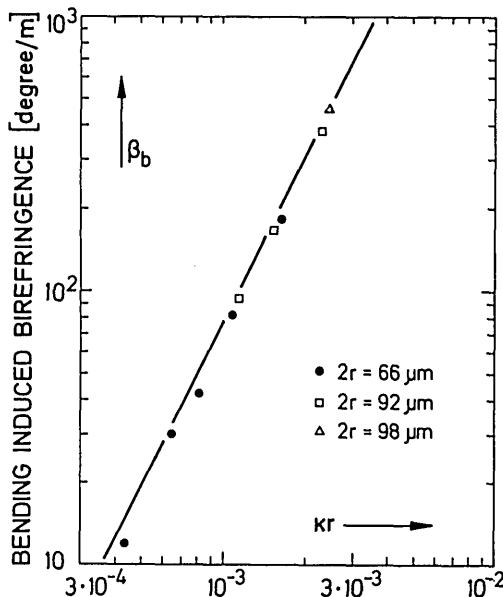


Fig. 2. Bending birefringence of single-mode silica fibers. The solid line represents the calculated birefringence [Eq. (5)]. The points are measurements at 0.633 and $0.676 \mu\text{m}$ using three fibers of different origins.

line representing Eq. (5). The birefringence reported in Ref. 4 seems to disagree with Eq. (5), but actually the agreement is excellent if the correct⁸ fiber diameter, $115 \mu\text{m}$, is taken into account.

The negative sign in Eq. (5) indicates that the fast axis of the bending birefringence coincides with the radius of curvature. The slow axis is normal to the plane of curvature. These directions agree with our observations.

In order to illustrate the absolute magnitude of the bending-induced birefringence, we consider as an example a fiber of $2r = 100\text{-}\mu\text{m}$ diameter coiled on a drum of $2R = 4\text{-cm}$ diameter. With $\kappa = 0.5 \text{ cm}^{-1}$, we find that $\beta_b = 481 \text{ deg/m}$, corresponding to a polarization beat length of 75 cm . Thus we see that for small coils the bending-induced birefringence becomes comparable to the intrinsic birefringence of typical present-day single-mode fibers.

In conclusion, we have explained the bending-induced birefringence as a second-order stress effect and have calculated and measured its magnitude. The result is useful in the design of fiber-optical interferometers.

Appendix

In the limit $\kappa r \ll 1$, we may calculate the lateral stress distribution as a perturbation $\sigma^{(1)}(x) + \sigma^{(2)}(x, y)$ superimposed upon the dominant stress $\sigma_z^{(0)} = \kappa E x$. In order to determine this perturbation from Eq. (2), we ignore the curved geometry and integrate Eq. (2) in Cartesian coordinates x, y, z . From ordinary first-order elastic theory,⁵ we find that Eq. (3) describes exactly the distribution $\sigma_x^{(1)}$ in a (bent) slab of thickness $2r$ that extends infinitely along the y and z directions and is subject to the constraints $\epsilon_y = \epsilon_z = 0$. By symmetry, no shear stress exists in such a slab, and $\sigma_y^{(1)} = \sigma_x^{(1)}\nu/(1 - \nu)$. This field $\sigma^{(1)}$ serves as the particular integral⁵ of the inhomogeneous Eq. (2).

This first perturbation $\sigma^{(1)}$ acts on each element of the fiber surface $x^2 + y^2 = r^2$ with an elastic force $\sigma_n^{(1)}$ whose Cartesian components are combined in complex form⁵ $F(\alpha) = \sigma_{nx}^{(1)} + i\sigma_{ny}^{(1)}$. Here α is the azimuth. We now imagine that the fiber is cut out of the slab as indicated in Fig. 1(c). Removal of the shaded material allows the fiber surface to relax into a state in which there is no normal surface stress. The stress field $\sigma^{(2)}(x, y)$ resulting from that relaxation is calculated by assuming a force $-F(\alpha)$ to act on the surface of an originally unstressed fiber. The total lateral stress in the bent fiber is the sum $\sigma^{(1)} + \sigma^{(2)}$. Actually, we require only the stress difference $\delta\sigma = \sigma_x(0) - \sigma_y(0)$ on the fiber axis. We obtain $\delta\sigma^{(1)} + \delta\sigma^{(2)} = -\kappa^2 r^2 E/2$ and, therefore, exactly Eq. (4), if we calculate $\delta\sigma^{(2)}$ by Muskhelishvili's method of complex integration.⁵ With $f(\alpha) = i \int F(\alpha) d\alpha$ and $\zeta = \exp(i\alpha)$ in that method,

$$f(\alpha) = -\kappa^2 r^2 (E/6) [i \sin^3 \alpha - \nu(1 - \nu)^{-1} \times (3 \cos \alpha - \cos^3 \alpha)], \quad (6)$$

$$\delta\sigma^{(2)} = (i/\pi) \oint \bar{f} \zeta^{-2} - 3f \zeta^{-4} d\zeta, \quad (7)$$

$$= -\kappa^2 r^2 (E/2)\nu(1 - \nu)^{-1}. \quad (8)$$

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